# Build your own

# AI Agent

## Microsoft Agent Framework & AG-UI

Medhat Elmasry & Jasmin Goh

# **AG-UI**: The Agent-User Interaction Protocol

**AG-UI** is an event-based protocol, created in May 2025, for **connecting AI agents to user-facing applications**.

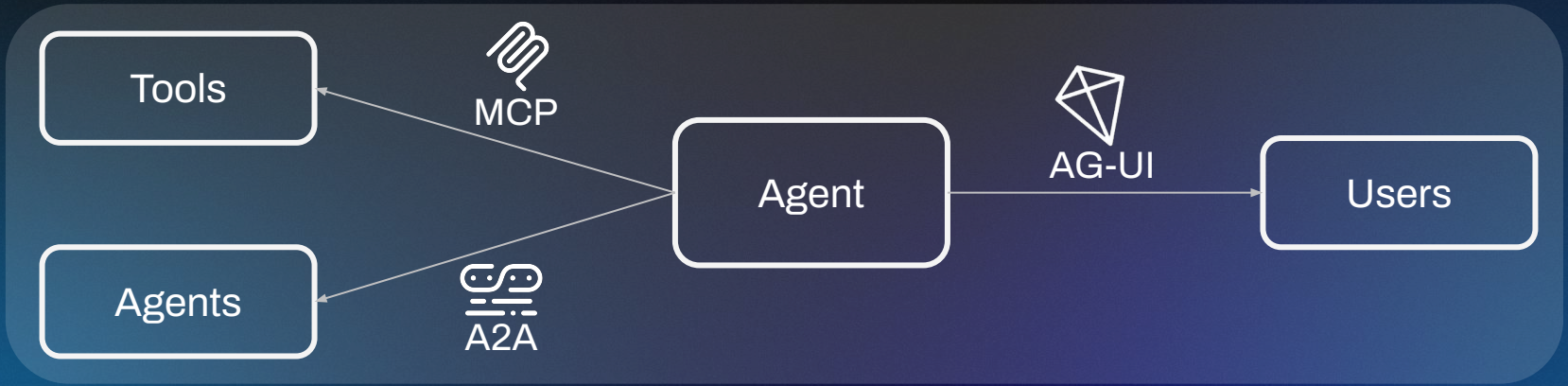
It acts as a **bi-directional bridge** between:

- Your agentic backend
- Your user-facing frontend

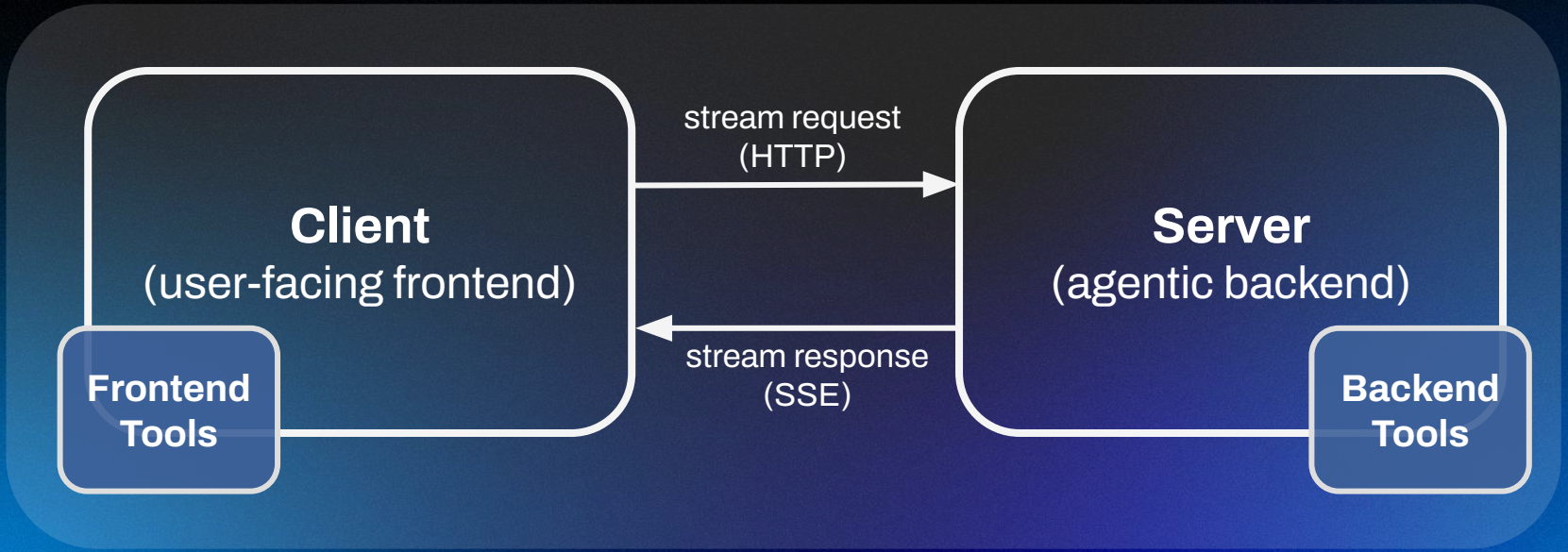
With AG-UI, you can **avoid complex ad-hoc wiring** for agentic features.

# The Agent Protocol Stack

- **MCP** standardizes agent-tool integration
- **A2A** enables agent-to-agent communication
- **AG-UI** bridges agents and users



# AG-UI Architecture



# Why AG-UI?

1. **Remote Agent Hosting**
   - Agents can be deployed as services **accessible by multiple clients** (web, mobile)
2. **Real-time Streaming**
   - Agent **responses can be streamed** using Server-Sent Events (SSE)
3. **Backend and Frontend Tools**
   - **Results** from backend tools are **automatically streamed to client**
   - **UI components can be modified** via frontend tools
4. **Human-in-the-Loop**
   - **User confirmation can be requested** before tool execution

# AG-UI Blazor demo

# Workshop

You'll be building a **console app** with a client-server architecture

1. **Server setup**
   Build an AG-UI server to host your AI agent
2. **Client setup**
   Create a UI that streams responses from the agent
3. **Tools**
   Add tools the agent can call
4. **Human-in-the-Loop**
   Add tools that require user approval to proceed

# Workshop: 1. Server Setup

## Build an AG-UI server to host your AI agent

By the end of this, you'll have:

- A running AG-UI server
- An AI agent hosted on the server
- An HTTP endpoint listening for requests

# Workshop: 2. Client Setup

## Create a UI that streams responses from the agent

By the end of this, you'll have:

- A client connected to the AG-UI server
- User input sent to the agent
- Agent responses streamed and displayed in the UI

# Workshop: 3. Tools

## Add tools your agent can call to perform tasks

By the end of this, you'll have:

- A backend tool that runs on the server
- A frontend tool that runs on the client
- An agent that decides when to call each tool

# Workshop: 4. Human-in-the-Loop

## Add tools that require user approval to proceed

By the end of this, you'll have:

- A tool that requests user approval
- An agent that pauses while waiting for input
- The ability to approve or reject action and resume execution

# Time to build!

## Follow the steps on https://bit.ly/netbc-agui:

0. **GitHub Token**: Get your GitHub PAT (if you don't have it yet)
1. **Server setup**: Build an server to host your agent
2. **Client setup**: Create a UI to streams agent response
3. **Tools**: Add tools the agent can call
4. **Human-in-the-Loop**: Add tools that require user approval

### Stuck?

Call Medhat or Jasmin!

We'll be going around the room making sure you're good

### Done…?

Build you own tools!

Connect to an API, analyze uploaded CSV, approve recipe ideas…

# Resources

- Our GitHub repo: https://bit.ly/netbc-agui
- AG-UI:
  - https://docs.ag-ui.com
- AI Agent protocols:
  - https://www.ibm.com/think/topics/ai-agent-protocols
  - https://modelcontextprotocol.io
  - https://a2a-protocol.org
- AG-UI Integration with Microsoft Agent Framework: https://learn.microsoft.com/en-us/agent-framework/integrations/ag-ui
- Building an AG-UI project: https://youtu.be/tDQc6lZUbYc?si=Uc4TE2Sn_BREIPtp

# Thank you

Take what you learned, build something yours, and come tell us about it at our next event!

**Questions?**

Medhat Elmasry
medhat@elmasry.ca

Jasmin Goh
hello.jasmin.noe@gmail.com